

Review

Diagnosis of postoperative arrhythmias following paediatric cardiac surgery

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SUMMARY

Arrhythmias are commonly encountered in the paediatric intensive care unit setting, most frequently in the setting of postoperative congenital heart disease. Postoperative arrhythmias are an important cause of morbidity in children in the postoperative period following cardiac surgery for congenital cardiac lesions. It is important for all paediatric critical care physicians involved in the care of these children to understand the potential mechanisms involved and how to make an accurate diagnosis. The existing literature has focused on small groups and specific arrhythmias. There is a paucity of literature to guide the clinician in approaching arrhythmias in the paediatric intensive care unit setting. Our objective was to review the recognition and diagnosis of paediatric arrhythmias in the postoperative period following congenital cardiac surgery. Timely and accurate identification of the rhythm disturbance is mandatory and allows for the institution of effective, rhythm specific management strategies.

Key Words: temporary pacemaker, congenital heart disease, arrhythmia, postoperative, children

The prevalence of postoperative arrhythmias varies between 15 and 48% following surgical repair of congenital heart disease (CHD)¹⁻⁵. Risk factors include lower age, prolonged cardiopulmonary bypass, long aortic cross-clamp time, deep hypothermia and circulatory arrest. Regardless of the underlying aetiology, prompt diagnosis followed by appropriate management is imperative in the paediatric intensive care unit (PICU).

Postoperative arrhythmias cause significant morbidity and are associated with a higher hospital mortality^{2,3}. Bradyarrhythmias are more frequent than tachyarrhythmias; supraventricular tachycardias occur more frequently than ventricular tachycardias in the paediatric setting^{1,2,4,5}. The inconsistencies in

the techniques used for diagnosis and definitions used to describe arrhythmias lead to inconsistent reporting, diagnosis and management strategies in the postoperative patient⁶. The aims of this review are:

1. To describe the aetiological factors which predispose the child with CHD to arrhythmias in the PICU.
2. To provide a practical classification of arrhythmia mechanisms for the bedside clinician in the PICU.
3. To provide a practical outline of the techniques for prompt diagnosis of the common arrhythmias for the bedside clinician in the PICU.
4. To provide specific examples and practical guidelines for diagnosing common arrhythmias in the PICU.

THE AETIOLOGY OF ARRHYTHMIAS IN THE POSTOPERATIVE CARDIAC SURGERY PATIENT

The electrical events of the cardiac cycle depend on specialised conduction tissue and intercellular electrical connectivity to optimise cardiac filling and output. The co-ordinated rhythmic contraction of the heart begins with spontaneous depolarisation of cells in the sinoatrial node due to diastolic depolarisation and calcium cycling^{7,8}. The electrical

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wavefront spreads from the sinoatrial node to the surrounding atrial myocardium, causing atrial contraction. The atrioventricular node (AVN) then serves as the electrical junction between the atria and the ventricles. The existence of internodal pathways has been debated^{9,10}. The AVN is intended to be the only electrical connection between atrial and ventricular myocardium. Accessory connections can exist which allow conduction to proceed antegradely, retrogradely or in both directions between atrium and ventricle. The specialised structure of the AVN allows for diastolic filling of the ventricles before the His-Purkinje system activates the ventricular myocardium. The conduction system is designed to provide optimal synchronised contraction of the atria prior to ventricular contraction, known as atrioventricular (AV) synchrony. As the atrial and ventricular cells become electrically activated, calcium influx prompts myocyte contraction. Repolarisation is an active process with restitution of the membrane potential in anticipation of the next electrical stimulus. Carefully regulated acid-base status, electrolytes and temperature are necessary to maintain normal cardiac electrical activity. Therefore, cardiac surgery

can affect the heart either at a macroscopic level or at a cellular level to promote arrhythmias.

Mechanisms of arrhythmias

In the clinical setting, arrhythmias are classified, based on the patient status as 'unstable' or 'stable'. This informs of the urgency for therapy. Beyond this, a mechanistic classification allows for a better understanding of the relationship between the underlying substrate and the therapy. Despite the multitude of factors which can cause arrhythmias, the mechanism can be categorised as shown in Table 1.

Automaticity refers to the ability of cardiac cells to depolarise spontaneously. Under normal circumstances, the greatest automaticity is in the cells of the sinoatrial node, followed by the cells of the AVN. Re-entrant circuits require two pathways capable of conduction with unilateral conduction delay in one limb and an intervening conduction barrier (Figure 1). Re-entry may involve a macro-re-entrant circuit, such as occurs with an accessory pathway. Alternately, re-entry can involve a micro-re-entrant circuit as might be seen around a small area of scarring¹¹. Examples of re-entrant

TABLE 1
Common postoperative arrhythmias based on mechanism and substrate

Mechanism		Anatomic substrate	Arrhythmia	Rationale
Automaticity	Decreased	Bidirectional Glenn, Fontan	Sinus node dysfunction	Previous surgery in vicinity of SAN ¹²⁻¹⁴
	Increased	All patients	Sinus tachycardia	Endogenous catecholamines, metabolic state, medications all can increase sinus rate
	Increased	Surgery near AVN (VSD, AVSD, ASO)	Junctional ectopic tachycardia	AVN automaticity enhanced by trauma and endogenous catecholamines ¹⁵
	Increased	RAI	Atrial tachycardia	Accessory sinus nodes ^{16,17}
Circuit	Re-entrant	Ebstein's	AVRT	Accessory pathways in 20-30% ¹⁸
	Re-entrant	HLHS	AVNRT	Dual AV node physiology common in left heart obstruction ¹⁹
	Re-entrant	Tetralogy of Fallot	Ventricular tachycardia	Right outflow tract conduction barriers is substrate for VT ²⁰
	Re-entrant	LTGA	AVRT	Same as Ebstein's for systemic AV valve ²¹
	Re-entrant	Classic Fontan	IART	Dilated atrium with scar and conduction barrier ²²
	Re-entrant	Right atrial dilatation	Atrial flutter	Allows use of right atrium as macro-re-entrant pathway ²³
	Re-entrant	Isomerism	Atypical AVNRT	Dual AVN ("Monckeberg sling") ^{24,25}
	Block	LTGA, LAI	Complete AV block	Abnormal AVN or rudimentary AVN ²⁶
	Triggered activity	Acquired QT prolongation	Torsade de pointes	Impaired repolarisation leads to after depolarisations ²⁷

SAN=sinoatrial node, AVN=atrioventricular node, VSD=ventricular septal defect, AVSD=atrioventricular septal defect, ASO=atrial switch operation, RAI=right atrial isomerism, AVRT=atrioventricular re-entrant tachycardia, HLHS=hypoplastic left heart syndrome, AVNRT=atrioventricular node re-entrant tachycardia, AV=atrioventricular, LTGA=levo-transposition of the great arteries, VT=ventricular tachycardia, IART=intra-atrial re-entrant tachycardia, LAI=left atrial isomerism.



FIGURE 1: A re-entrant circuit. There are two limbs with different refractory periods with an intervening area of insulation.

tachycardias are listed in Table 1. A third mechanism of arrhythmias is triggered activity, seen more commonly in certain arrhythmia substrates such as long QT syndrome and digoxin toxicity¹²⁻¹⁴.

Anatomical and surgical factors

In children with CHD, preoperative and intraoperative factors contribute to the development of postoperative rhythm disturbances. The underlying anatomy is an important predictor of postoperative arrhythmias. Several congenital heart defects are associated with arrhythmia substrates (Table 1)^{11,15-22}.

Chronic volume or pressure loading leads to chamber hypertrophy and fibrosis. This and previous palliative or corrective surgery may result in anatomic barriers to conduction and scarring. This sets up re-entrant areas around areas of diminished electrical activity, such as is seen in incisional tachycardias, where a circuit exists around the non-conducting scar²³⁻²⁵.

During the conduct of surgery itself, bypass cannulation is near the sinus node, which may impair sinus node automaticity. Sinus node dysfunction can be seen acutely and chronically following open heart surgery^{11,26}. Many procedures for the correction of CHD involve the manipulation of tissues in the vicinity of the AVN (e.g. ventricular septal defects,

AV septal defects, arterial switch). Normal AVN conduction may be affected by tissue trauma related oedema or by direct injury from suture lines²⁶. Junctional ectopic tachycardia (JET) – the most common tachycardia in the postoperative CHD patient – occurs when surgery in the vicinity of the AVN results in local changes that appear to enhance automaticity²⁷⁻²⁹.

Electrolytes and metabolic state

Disturbances of electrolytes are common following open-heart surgery and can affect rate, rhythm and automaticity^{30,31}. The most common abnormalities affect potassium, magnesium and acid-base balance. Frequent monitoring of electrolytes and blood gases is imperative. Preoperative diuresis can predispose to sodium, potassium and magnesium depletion. Postoperative renal impairment, bleeding, blood transfusion and diuretic requirements can aggravate these electrolyte imbalances. Potassium is important for maintaining a stable cell membrane potential. Hypokalaemia (<3 mmol/l) increases the cell membrane potential and is associated with delayed conduction causing atrial and ventricular ectopy, atrial tachycardia and AV block. Ventricular tachycardia and fibrillation are uncommon. Hyperkalaemia is associated with sinus arrest and PR interval changes, although AV block is uncommon^{30,32}. Hyperkalaemia accompanies oliguria or renal failure and rapid blood transfusion for postoperative bleeding. Severe bradycardia, idioventricular rhythms, ventricular arrhythmias and asystole are seen at higher serum levels^{32,33}. Magnesium has many important functions on the cell membrane. It is a critical cofactor for Na⁺-K⁺-ATPase, a key transporter in maintaining the cell membrane potential, a regulator of some cell membrane K⁺ channels and calcium antagonist. Hypomagnesaemia is often associated with hypokalaemia. Ventricular ectopy and ventricular fibrillation has been described³⁴. Magnesium supplementation to maintain a normal serum magnesium have been proposed as a therapy for JET²⁹. Hypermagnesaemia is uncommon unless iatrogenic but can be associated with delays in AV and intraventricular conduction.

Management of arterial pH is controversial. Metabolic acidosis resulting from hypoperfusion is best managed by treating the underlying cause and optimising oxygen delivery rather than prescribing bicarbonate. In situations where oxygen delivery is severely impaired and further incremental improvement requires time, some clinicians prescribe bicarbonate to correct the metabolic acidosis although the supportive evidence of its

effect on long term outcome is lacking. Caution with bicarbonate use in the situation of low cardiac output is warranted to prevent severe hypokalaemia and associated ventricular arrhythmias. Although commonly prescribed for hyperkalaemia, there is little supportive evidence for its use in this situation³⁵.

Fever is common in the postoperative period and reflects the inflammatory state induced by cardiopulmonary bypass. Fever increases the endogenous catecholamines and can accelerate automatic tachycardias.

Medications

The spectrum of medications that can predispose patients to postoperative arrhythmias is extensive. The most frequent medications prescribed in children with heart disease are diuretics. Diuretics commonly cause both acute extracellular and chronic intracellular electrolyte abnormalities and set the stage for arrhythmias. Drug interactions with antiarrhythmic agents in combination with electrolyte disorders are of particular importance. Atrioventricular block is the most common side-effect of these agents in the postoperative period^{36,37}. Amiodarone is the most common antiarrhythmic used in postoperative paediatric cardiac intensive care³⁸. It is safe and effective but has interactions with a variety of medications; specifically amiodarone can lead to decreased clearance of digoxin resulting in toxicity (and atrioventricular block). In addition, all antiarrhythmic agents have the propensity to cause arrhythmias (proarrhythmia) due to direct or indirect effects on membrane ion conductance, which in turn alters the cardiac action potential.

Other medication-induced mechanisms of postoperative arrhythmia include QT prolongation, a disorder of prolonged ventricular repolarisation which needs to be recognised in the acute postoperative setting (www.qtdrugs.org). The more common medications which can prolong the QT interval include antiarrhythmic medications, macrolide antibiotics and antifungals. Calcium influx into cells is also a common pathway in the genesis of arrhythmias, particularly in automatic arrhythmias and those caused by triggered activity^{30,31,39}. Exogenous inotropes (catecholamines, phosphodiesterase inhibitors) are associated with an increase in arrhythmogenesis, presumably through their actions in increasing intracellular calcium. Dopamine in particular has been associated with JET⁴⁰. Beta-blockers can also be implicated if abruptly withdrawn preoperatively, leading to a postoperative sinus tachycardia and hypertension; postoperative use such as in coarctation repair may be associated with bradycardia or AV block; or proarrhythmia.

IMPACT OF ARRHYTHMIAS IN THE POSTOPERATIVE SETTING

The common reasons for haemodynamic compromise with any arrhythmia in the child are related to rate (either too slow or too fast) and loss of AV synchrony³⁰. Cardiac output is determined by the stroke volume and the heart rate. Blood pressure is determined by the systemic vascular resistance and the cardiac output. In children, the primary way to increase cardiac output is to increase the heart rate. Therefore, arrhythmias which result in a low heart rate often lead to low cardiac output. Tachycardia can cause haemodynamic compromise by decreasing diastolic filling time and stroke volume, increasing myocardial oxygen consumption and impairing coronary perfusion. Blood pressure is maintained in both cases by increasing the systemic vascular resistance and therefore afterload.

Loss of AV synchrony also reduces cardiac filling. In a compliant heart, most of the ventricular filling occurs during the rapid early filling phase of ventricular diastole. Preoperative pressure and volume loading increase the role that atrial contraction plays in ventricular filling and cardiac output. While the preoperative anatomy may contribute to ventricular hypertrophy, thereby impairing diastolic compliance, cardiac surgery exacerbates this by increasing myocardial oedema and dysfunction. An increase in heart rate postoperatively results in a corresponding decrease in filling time. Therefore, in the acute postoperative period, atrioventricular synchrony plays a much greater role than in the same child preoperatively. Fortunately, most arrhythmias encountered in the postoperative period are transient in nature^{41,42}. As well, the effects of cardiopulmonary bypass are self-limited and the need for AV synchrony and chronotropic competence decrease as the child improves. The most common sustained arrhythmias are bradycardic: usually sinus node dysfunction or complete heart block^{1,5}. These sometimes necessitate long-term pacemaker therapy⁴³.

CLINICAL APPROACH TO THE DIAGNOSIS OF AN ARRHYTHMIA

The first priority in the child with a postoperative arrhythmia is ensuring that haemodynamic stability is established. The 2005 Pediatric Advanced Life Support algorithms are a useful approach to the initial management of arrhythmias in a child⁴⁴. Accurate diagnosis and management of postoperative arrhythmias requires careful preoperative preparation and a team approach to care. A

thorough understanding of the patient's anatomy and haemodynamics is essential.

On arrival in the PICU following cardiac surgery, the stability of the patient's underlying rhythm and dependence upon pacemaker function is important handover information between the cardiac anaesthetist and PICU physician. A systematic approach to monitoring and diagnosis of rhythm abnormalities includes a rhythm strip printout and a 12-lead electrocardiogram to establish a template to compare QRS morphology if an arrhythmia develops, or make a diagnosis of an existing arrhythmia. The latest generation of patient monitors is capable of multichannel recording and retrospective full disclosure review of all electrocardiographic data and events.

Simultaneously observing the arterial pressure and waveform looking for beat-to-beat variability in the pulse pressure can aid in detecting arrhythmias and pulsus paradoxus. A regular, biphasic atrial pulse contour on the bedside monitor is suggestive of AV synchrony, whereas a monophasic pulse contour, marked irregularity or the presence of atrial cannon waves provides evidence for a lack of appropriate AV synchrony. This lack of synchrony may be due to AV dissociation, retrograde activation of atria or normal activation in the presence of extreme first-degree AV block. An echocardiogram should be urgently obtained in any postoperative patient with new onset arrhythmia to exclude a haemodynamic cause, such as residual outflow tract obstruction, tamponade due to blood or myocardial swelling or reduced function. Compromise of coronary perfusion, such as can be seen in arterial switch operations, in the retrogradely perfused coronaries post-Norwood palliation for hypoplastic left heart syndrome or any other procedure involving coronary manipulation (e.g. Ross procedure) can lead to sinus tachycardia changes or bradycardia due to compromise of coronary perfusion.

In the presence of atrial pacing leads, accurate diagnostic information can be obtained by obtaining a multi-lead atrial electrogram recording. The atrial electrogram allows for precise identification of the timing of atrial depolarisation in relation to the ventricular QRS complex. An atrial electrogram can be recorded using standard electrocardiogram (ECG) equipment in one of two ways. A unipolar atrial recording is obtained by connecting the limbs as usual and the atrial wires to two of the precordial leads (e.g. V1 and V2) and simultaneously recording concurrent limb leads on the ECG (Figure 2A). If this does not yield a

clear atrial electrogram, a bipolar atrial recording can be obtained by attaching the lower limb leads in the usual fashion and connecting the atrial epicardial wires to the right and left upper limb leads (Figure 2B). This generates a bipolar atrial electrogram in lead I and a 'hybrid' in leads II and III. This method has the disadvantage that concurrent surface limb leads are not available for simultaneous comparison. The resolution of atrial timing using either techniques of atrial electrocardiography is superior to that provided by the standard surface electrocardiogram.

In the case where atrial wires are not available, an oesophageal lead can be used to facilitate identifying the atrial depolarisation on the ECG. The oesophageal lead is passed through the nose or the mouth and positioned near the mid-oesophagus directly behind the left atrium. For the neonate or infant, optimal electrical signals are typically found at an insertion depth of 12 to 17 cm measured from the distal electrode to the nares or mouth⁴⁵. The oesophageal probe can be connected to the ECG machine in the same way that the atrial epicardial wires can be, allowing a recording of the atrial electrogram.

The relationship between the atrial activity and the ventricular activity may be difficult to ascertain in some cases of tachycardia. Adenosine administered during the recording of an atrial electrogram can be diagnostic by transiently blocking AVN conduction. It is important to be able to pace and cardiovert in this situation, as adenosine may result in atrial arrhythmias, in addition to the anticipated bradycardia⁴⁶.

Additional diagnostic information can be provided in patients with a tachycardia of unclear mechanism by a brief period of atrial overdrive capture that is abruptly terminated while observing the response. If the tachycardia is utilising a re-entrant circuit and the circuit can be penetrated by the external pacing stimuli, the tachycardia may terminate. If the tachycardia is automatic, it may be briefly suppressed before resuming. The role of overdrive pacing is discussed in detail elsewhere⁴⁷.

SPECIFIC ARRHYTHMIA EXAMPLES

The arrhythmias most commonly encountered in the postoperative CHD patient are sinus bradycardia (relative or absolute), complete AV block and JET. However, for completeness, other less common arrhythmias are also described below. Figure 3 provides a practical approach to the accurate diagnosis of postoperative arrhythmias.

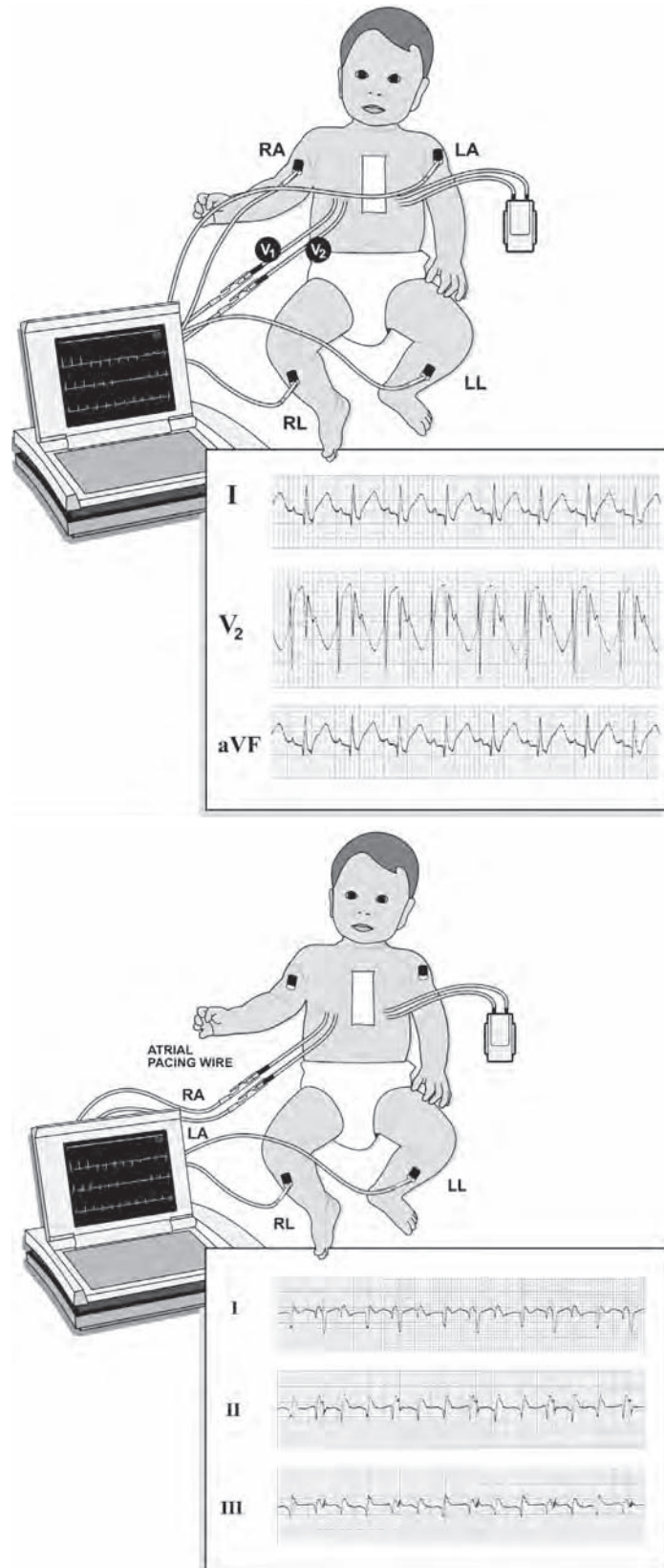
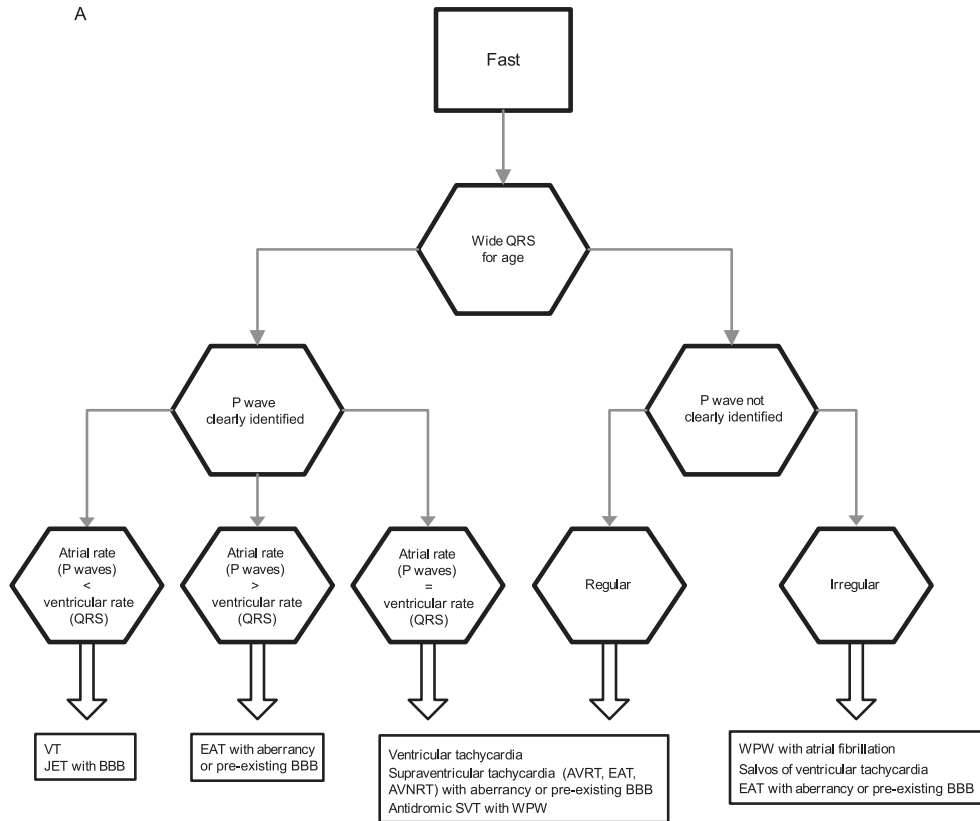
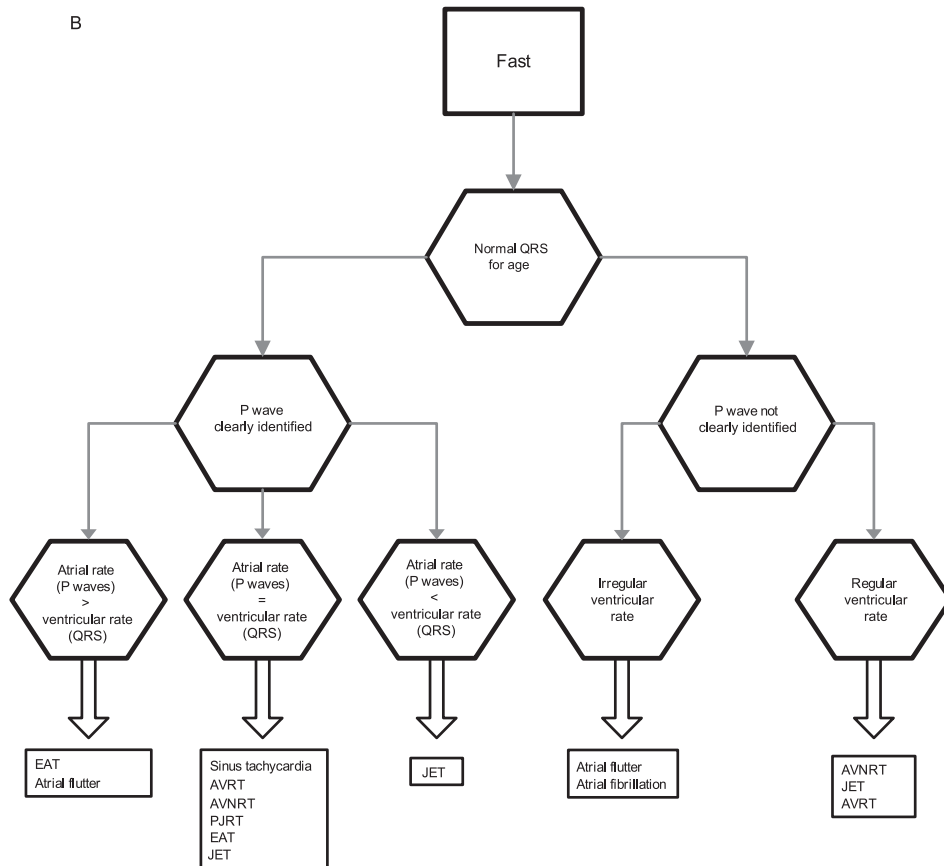


FIGURE 2: Illustration on how to perform a wire study. A) Unipolar setup: both atrial leads are connected to the unipolar precordial leads based on which leads the ECG machine can display concurrently with at least one limb lead. B) Bipolar setup: both atrial wires are connected to the right arm and left arm leads.

A



B



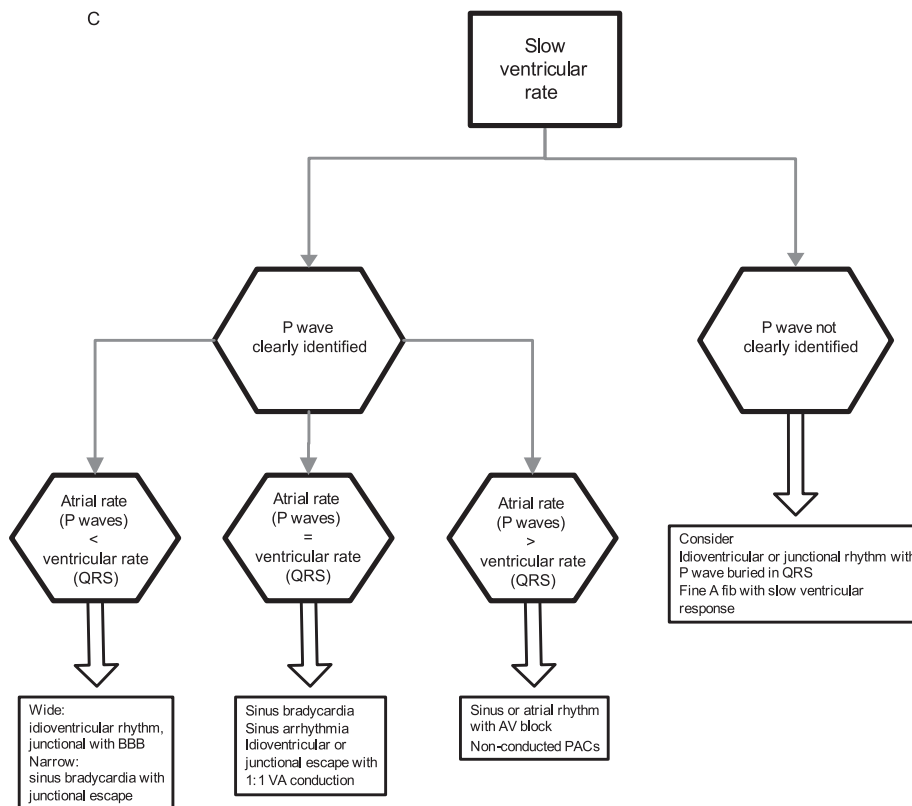


FIGURE 3: Classification of arrhythmias based on ECG diagnosis. EAT=ectopic atrial tachycardia, AVRT=atrioventricular reciprocating tachycardia, AVNRT=atrioventricular node re-entrant tachycardia, SVT=supraventricular tachycardia, WPW=Wolff-Parkinson-White, PJRT=permanent junctional reciprocating tachycardia, JET junctional ectopic tachycardia, PAC=premature atrial contraction, AV=atrioventricular block.

Bradyarrhythmias

Sinus bradycardia (Figure 4A) is diagnosed on the surface ECG or atrial electrogram as a P wave preceding a normal QRS complex at a rate that is slow for age and physiologic requirements. The threshold for diagnosing sinus bradycardia is dependent on the age of the patient and the clinical circumstances. A wide range of heart rate definitions have been given, some referring to sleeping rates from 24 hour Holter studies, and others to the lower rates from an awake, standard ECG. For example, a heart rate of 65 beats/minute in a sleeping, sedated three-year-old with normal haemodynamics is within normal; whereas if the child were agitated or unstable and on inotropic support, a rate of 90 beats/minute would indicate a relative bradycardia. Due to the difficulty in defining sinus bradycardia, the prevalence of sinus bradycardia is frequently not reported; however, in one study⁵ sinus bradycardia occurred in 46% of patients with arrhythmias. Whenever the ventricular rate appears to be lower than anticipated in a postoperative patient, it is important to exclude atrial tachycardias with 2:1 AV block, in which alternate P waves may be difficult to identify.

AV block is present when there is an abnormality of impulse propagation from atrium to ventricle. First-degree AV block (Figure 4B) is defined as a long conduction time from the onset of atrial depolarisation to ventricular depolarisation. This includes the trans-atrial conduction time and the AV nodal conduction delay. This is manifest as a P wave followed by a long PR interval before the QRS complex. The normal values are based on age and rate⁴⁸. For example, a PR interval of 170 ms in a teenager is normal, but not if the heart rate is 150 beats/minute. In severe cases, the long conduction time from atrium to ventricle may impede ventricular filling as atrial depolarisation occurs prior to ventricular relaxation. First-degree AV block is common in the presence of CHD and in children with electrolyte disturbances or receiving antiarrhythmics.

There are two types of second-degree AV block. The first is known as Mobitz Type I or Wenckebach and typically reflects conduction delay at the AVN. This is manifest by progressive lengthening of the PR interval, with eventual non-conduction of a P wave (Figure 4C). Type II second-degree block is usually representative of a more advanced

conduction abnormality of the His-Purkinje system. This is manifest by a constant PR interval with one or more P waves not conducting to the ventricle (Figure 4D). Second-degree block can result in inappropriately low ventricular heart rates. From published reports, the prevalence of second-degree AV block appears to be low in the postoperative period⁴². Second-degree block is a common cause of an irregular rhythm.

Third-degree AV block or complete heart block is present when impulses from the atrium are not conducted to the ventricles (Figure 4E). This is not synonymous with AV dissociation, which simply means that the atria and the ventricles are being driven by different pacemaking foci. For example, during JET with AV dissociation, the ventricle is being driven by a nodal focus, while the atria are driven by the sinus node. An appropriately

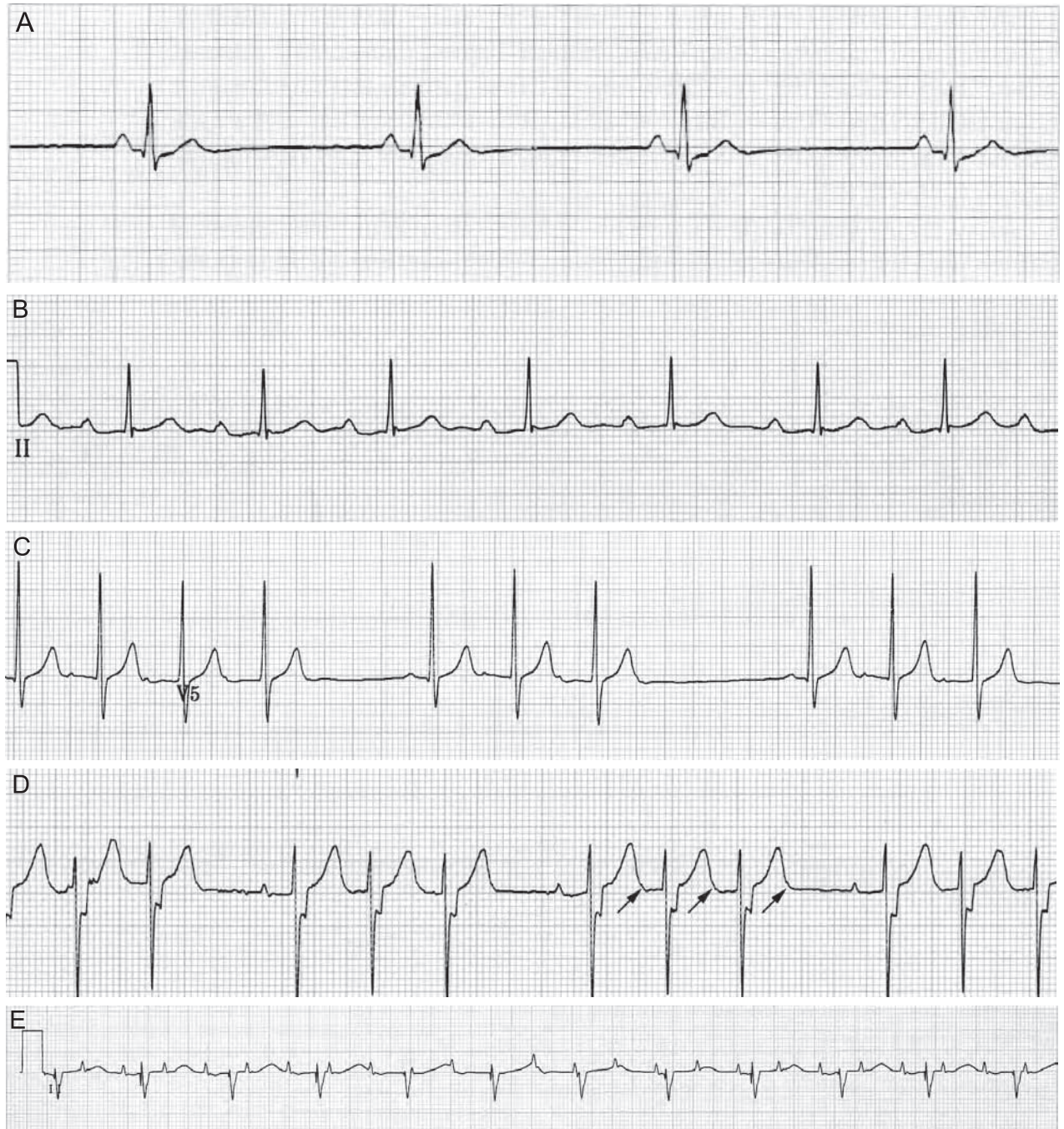


FIGURE 4: Bradycardia examples: A) sinus bradycardia, B) first-degree atrioventricular (AV) block, C) Mobitz I or Wenckebach AV block – note the progressing lengthening of the PR interval, D) Mobitz II AV block – the arrows indicate P waves, E) complete heart block.

timed sinus impulse can still be transmitted to the ventricles. Complete heart block is the most common postoperative arrhythmia requiring long-term therapy, and most commonly arises after surgery near the AV node (e.g. ventricular septal defect repair, atrioventricular septal defect repair, Tetralogy of Fallot repair)⁴⁹. The escape pacemaker can be AV nodal, from the His-Purkinje system, or ventricular, and the rate is usually too slow to maintain an adequate cardiac output in the postoperative setting. Without pacing, cardiac output is impaired by low heart rates and loss of AV synchrony. Spontaneous resolution is frequent and may occur in up to two-thirds of patients by nine days post-operatively⁵⁰. Late recurrence has been reported in as many as 5% of transient cases⁴³ and late recovery may occur in as many as 10% of patients who receive pacemakers⁵¹. It is important to document even transient complete heart block, as it has been reported to be a predictor of mortality in certain groups^{3,5}.

Tachyarrhythmias

Tachycardias comprise the most difficult arrhythmia to accurately diagnose and treat in the postoperative period. The most common tachyarrhythmias in the postoperative period result from enhanced automaticity. These include sinus tachycardia, JET and ectopic atrial tachycardia (EAT)^{39,52}. The ECG and clinical features that would suggest enhanced automaticity as the cause of tachycardia include:

- onset with steady warming-up (increase) of the rate (over 10 to 20 cardiac cycles),
- termination with cooling-down (slowing) of the rate,
- a variable rate that changes with manipulation of autonomic tone or temperature,
- refractoriness to electrical cardioversion or overdrive pacing manoeuvres (although they may be transiently suppressed with overdrive pacing).

Sinus tachycardia (Figure 5A) is the most common form of tachycardia and can be difficult to diagnose, particularly when there is an associated first-degree



FIGURE 5: Tachycardia examples: A) sinus tachycardia, B) junctional ectopic tachycardia wire study demonstrating ventriculoatrial dissociation, C) junctional ectopic tachycardia with ventriculoatrial dissociation,



FIGURE 5 (continued): D) ectopic atrial tachycardia – note the bizarre P waves and the Wenckebach conduction, E) atrial flutter – the arrows indicate P waves, F) permanent junctional reciprocating tachycardia – note the long RP interval.

AV block. In the absence of an associated first-degree AV block, the P wave is followed closely by a QRS, in a 1:1 relationship. The P wave morphology should be the same as the preoperative ECG, with a P wave axis consistent with origin at the superior vena cava-right atrial junction. The rate should be physiologic and does not usually exceed 220 beats/minute. In sinus tachycardia with first-degree block, the P wave may fall within the T wave and not be detected on surface ECG or be mistaken for a retrograde P wave. The wire study will demonstrate regular 1:1 relationship of the atrial impulse to the QRS with a relatively long PR interval for age and/or rate. Pacing at a rate just above the prevailing sinus rate should maintain 1:1 conduction, the same QRS morphology and PR interval. Sinus tachycardia also demonstrates a gradual warming-up and cooling-down to stimuli such as fever. A decrease in heart rate with optimising haemodynamics, controlling body temperature and providing adequate analgesia suggests an automatic focus, most commonly sinus tachycardia.

Postoperative JET (Figure 5B) is the most common postoperative malignant tachyarrhythmia and is caused by a focus of increased automaticity in the

AV node or His bundle^{39,53}. The morphology of the QRS complexes is usually the same as in normal sinus rhythm – either narrow complex or exhibiting pre-existent bundle branch block. The rhythm is usually incessant and, in cases with ventriculoatrial (VA) dissociation, the diagnosis is readily made (Figure 5C). The comparison to the sinus QRS is essential to exclude ventricular tachycardia (VT), particularly if the QRS is wide. It is not necessary to document VA dissociation as 1:1 retrograde conduction is well described, broadening the differential diagnosis to include sinus tachycardia with first-degree AV block or other tachycardias. The wire study can be used to exclude sinus tachycardia with first-degree block by careful comparison of RP and PR intervals during atrial and/or ventricular pacing (Figure 6). We recommend starting with atrial pacing and comparing the PR intervals between the intrinsic rhythm and the paced rhythm. Occasionally, ventricular pacing is diagnostic. For example, if the RP interval is significantly shorter during ventricular pacing than during tachycardia, the P wave is not likely to be retrograde and the diagnosis is likely to be sinus or ectopic atrial tachycardia. Pacing manoeuvres do not usually terminate JET; other

forms of supra-ventricular tachycardia (SVT) should be considered if this occurs.

As with other tachycardias, haemodynamic compromise attributable to JET is due to the increased heart rate and loss of AV synchrony with resultant decreased cardiac output. Accelerated junctional rhythm can be thought of as 'slow JET' and should be watched closely as it may gradually, or suddenly, increase in rate and become clinically significant.

EAT (Figure 5D) may be either paroxysmal or incessant. The characteristic feature of EAT is an inappropriately elevated atrial rate with a brief warming-up and cooling-down¹⁵. The QRS during EAT should be the same as during sinus rhythm, unless there is aberrancy. Most patients demonstrate first-degree AV block while a small proportion have second-degree block. EAT is often refractory to medical therapy and is not usually responsive to direct current cardioversion, but may be briefly suppressed by pacing at a faster rate. Diagnosis is determined by demonstrating an elevated atrial rate, occasionally with a non-sinus P wave axis or morphology. The wire study would show a rapid atrial rate with 1:1 conduction and first-degree block or an atrial rate in excess of the ventricular rate, when there is associated second-degree block. The most important role of the wire study is to differentiate EAT from atrial flutter with a P wave buried in the QRS or T wave.

This is important because flutter usually responds to overdrive pacing or cardioversion.

In contrast to automatic arrhythmias, the ECG and clinical features that would suggest re-entrant tachycardia include:

- abrupt onset and offset,
- abrupt termination with electrical cardioversion or overdrive burst pacing,
- ability to initiate and terminate with appropriately timed extra stimuli,
- minimal beat-to-beat variation.

Atrial flutter or intra-atrial re-entrant tachycardia can be seen in patients with CHD acutely and chronically. The latter refers to a tachycardia due to a macro re-entrant circuit which is distinct from the typical flutter circuit seen in the absence of structural heart disease. Intra-atrial re-entrant tachycardia can occur around incision sites from cannulation from bypass or around other barriers to conduction. Therefore, every patient undergoing open heart surgery has the potential substrate for intra-atrial re-entrant tachycardia. Due to rapid atrial rates, it is important to 'look for the missing P wave' in any patient with an unexplained persistent tachycardia (Figure 5E).

In patients with accessory pathways or dual AVN physiology, one can see re-entrant tachycardias in the postoperative period. The QRS during these forms

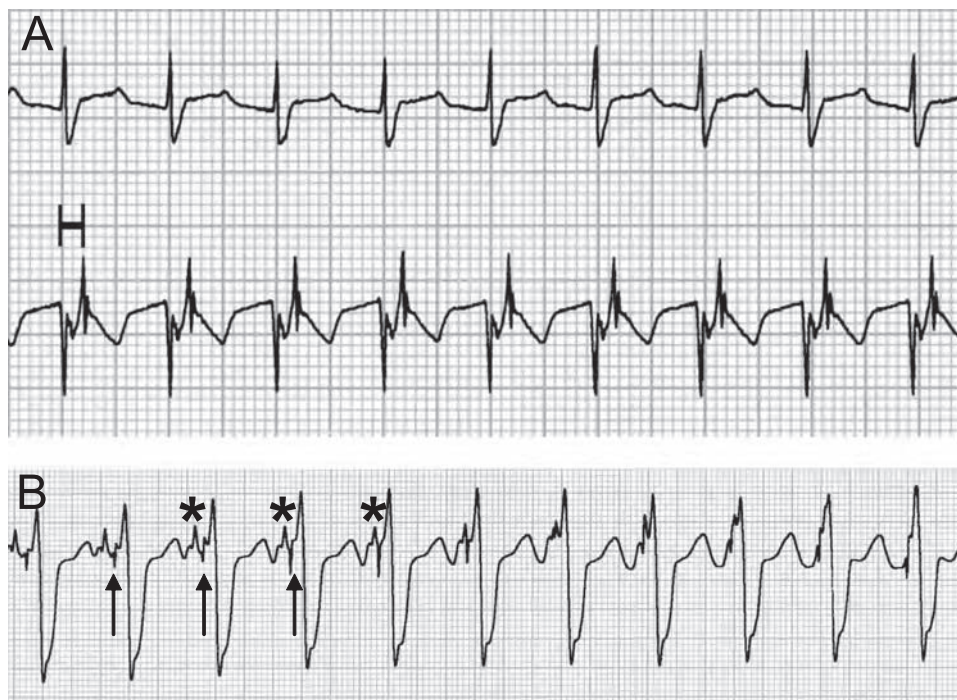


FIGURE 6: Atrial wire study during (A) intrinsic rhythm shows 1:1 ventriculoatrial relationship with RP interval of ~80 ms. Although one assumes this is retrograde conduction, comparison with (B) ventricular pacing demonstrates a long VA with VA dissociation, indicating that the intrinsic rhythm represents an atrial rhythm with antegrade conduction with long PR interval. Arrows indicate ventricular pacing spike; asterisks denote P waves.

of SVT should be the same as during sinus rhythm, unless there is aberrancy. An atrial electrogram during AV reciprocating tachycardia and AVN re-entrant tachycardia demonstrates a regular relationship of the atrial impulse with the QRS and a short RP interval. When a long RP interval (RP interval greater than PR interval) is observed, EAT and permanent junctional reciprocating tachycardia should be considered. Permanent junctional reciprocating tachycardia is an uncommon form of tachycardia but is due to a slowly conducting accessory pathway, resulting in a long RP interval, making the differentiation from EAT challenging^{1,2,4} (Figure 5D). Overall, SVT, other than sinus tachycardia and JET, has a low prevalence and occurs in less than 5% of postoperative patients and accounts for less than 20% of the arrhythmias documented^{1,2,4,5}.

The administration of adenosine induces block in the circuit, usually within the AV node. Termination of an arrhythmia by adenosine is essentially diagnostic of a re-entrant arrhythmia that uses the AV node as an essential part of the circuit (AVN re-entrant tachycardia, AV reciprocating tachycardia). A rare exception is the occasional adenosine sensitive atrial 'ectopic' tachycardia or ventricular tachycardia⁴⁶.

Ventricular tachycardia is an uncommon arrhythmia in children postoperatively, and occurs less than 2% of the time⁵³. It is important to bear in mind that in the PICU, VT can look relatively narrow (QRS ~100 ms) due to patient age, origin of the focus near the His-Purkinje system or rapidity of conduction. Asymptomatic premature ventricular contractions are not infrequent and have been seen in as many as 11% of all postoperative patients¹⁻⁴. ECG criteria for diagnosis of VT include:

- AV dissociation (present in the majority of VT) – the absence of VA dissociation is not helpful, because VT may happen with 1:1 VA conduction.
- widened or change in morphology of QRS complex from baseline. The QRS complex in neonates and infants may appear to be relatively narrow but of different morphology compared with sinus.
- intermittent fusion or capture beats.
- narrowing of QRS during atrial pacing at a faster rate.

In the presence of surgically-induced bundle branch block, it can be difficult to distinguish VT from SVT in children. A wide QRS can also be due to aberrancy, medication effect, pre-existing conduction abnormalities, an antegrade conducting

accessory pathway or a paced rhythm with tracking of a supraventricular arrhythmia. Distinguishing VT from SVT is clearly critical and, importantly, the rate and stability of the patient are not helpful in differentiating SVT with a wide QRS from VT in children. If the patient is stable, different manoeuvres can be used in an attempt to accurately diagnose the tachyarrhythmia. These include accurate recording of atrial electrograms and response to vagal manoeuvres or adenosine. Atrial pacing resulting in a narrowing of the QRS complex is suggestive that the arrhythmia is VT. In the unstable patient in whom the mechanism remains unclear, Pediatric Advanced Life Support guidelines should be followed and treatment for VT should be given⁴⁴.

PREVENTION OF ARRHYTHMIAS

Earlier repair may reduce the effects of long-standing volume and pressure load, and late onset arrhythmias. For the critical care physician caring for children in the immediate post cardiac surgery period, the key to preventing postoperative arrhythmias is anticipation. This requires a thorough understanding of the preoperative anatomy and conduct of cardiopulmonary bypass, the surgical repair and surgical factors contributing to arrhythmias, and postoperative factors that exacerbate or cause arrhythmias. Most children with open heart surgery will return to the PICU with atrial and ventricular pacing wires. These wires should be attached to the temporary pacemaker and the pacemaker function checked. Fever exacerbates automatic foci so maintenance of normothermia is imperative. Inotropes should be continuously reassessed and minimised. Optimising lung recruitment and careful monitoring of arterial blood gases to minimise pulmonary vascular resistance is critical to prevent pulmonary hypertension and right atrial and ventricular distension. Maintaining normal electrolytes is most safely managed with set nursing orders for prescribing replacement doses of potassium or magnesium. In the setting of reduced renal function, early institution of peritoneal dialysis will prevent hyperkalaemia and hypermagnesaemia.

SUMMARY

Abnormalities in cardiac rhythm are common following surgery for congenital heart disease and are an important cause of impaired cardiac output in this setting. Timely and accurate identification of the rhythm disturbance is mandatory, and allows for the institution of effective, rhythm-specific management strategies in these patients.

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